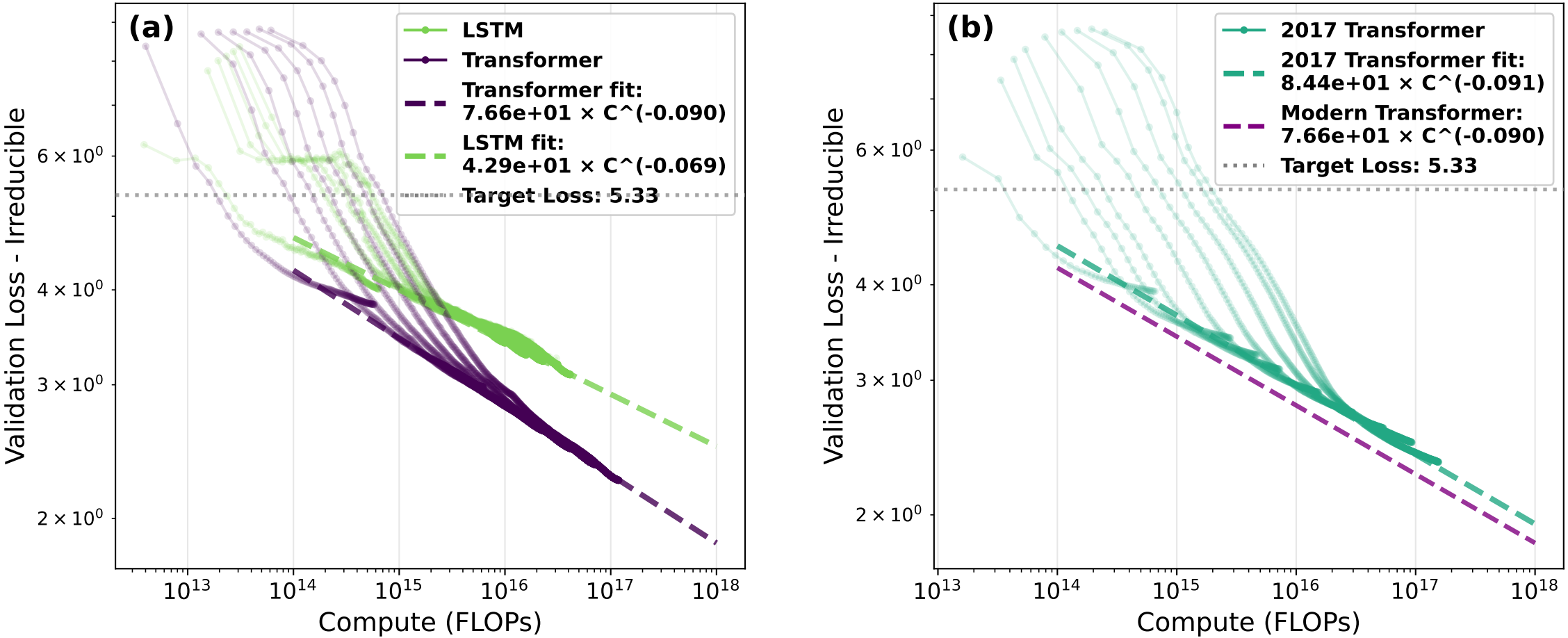Scaling Analysis: Architecture Comparison